



ARES

ADVANCED ROTORCRAFT ENGINEERING SYSTEM

Politecnico di Torino



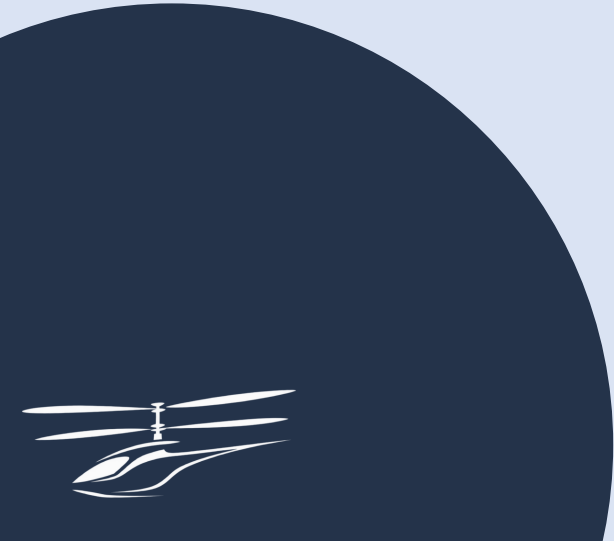
Politecnico
di Torino

Who we are

ARES is a student team at Politecnico di Torino founded to design and develop a modular, autonomous coaxial helicopter platform.

The team brings together expertise in aerodynamics, propulsion, structures, avionics and systems management to cover the entire development cycle of a rotorcraft, from concept to flying platform.

The project was born as a multi-year initiative, designed to grow competition after competition towards a complete physical prototype.



Why a Helicopter and Why Autonomous

HELICOPTER vs. MULTIROTOR DRONE

1

Efficiency & payload

A large rotor disk delivers a far better payload-to-power ratio and much longer endurance than a multirotor; essential for heavy sensors and wide-area search.

2

Stability in wind

Higher rotor inertia holds a steady hover through the gusts typical of urban canyons and off-shore sites, where small multirotors struggle.

3

Compact, simple airframe

The coaxial layout keeps a drone-like footprint and mechanical simplicity, no tail rotor, while keeping helicopter-level aerodynamic efficiency.

WHY AUTONOMOUS

1

Speed of response

No wait for a rated pilot: an autonomous unit can launch within minutes of an alert, when every minute changes the outcome.

2

Crew safety

Keeps people out of collapsed structures, wildfire fronts and GPS/comms-denied zones, flying the exposure instead of a person.

3

Scalable, repeatable coverage

Multiple units can fly precise patterns around the clock, without pilot fatigue or availability limits.





Mission

We design an **autonomous modular helicopter** in a coaxial configuration: the absence of a tail rotor frees up the space around the fuselage, facilitating access to and handling of the modules. The goal is a platform in which the mission module can be interchanged according to operational needs.



Sustainability

We focus on **concrete environmental sustainability**, made possible by full-electric propulsion: zero local emissions, low acoustic impact and full compatibility with existing urban infrastructure. The absence of combustion reduces vibrations, emissions and maintenance complexity, making the aircraft easier to integrate into urban and environmentally sensitive contexts.

Potential project concept



https://www.miquadra.it/works/industrial/eumac_09/

EUMAC 09 Ultralight multirole helicopter

An ultralight helicopter suitable for everything from recreational flying to emergency medical services, territorial surveillance, and civil protection. The core concepts are high functionality and payload versatility combined with an innovative and highly recognizable design, achieved through simplified production methods at reduced costs.

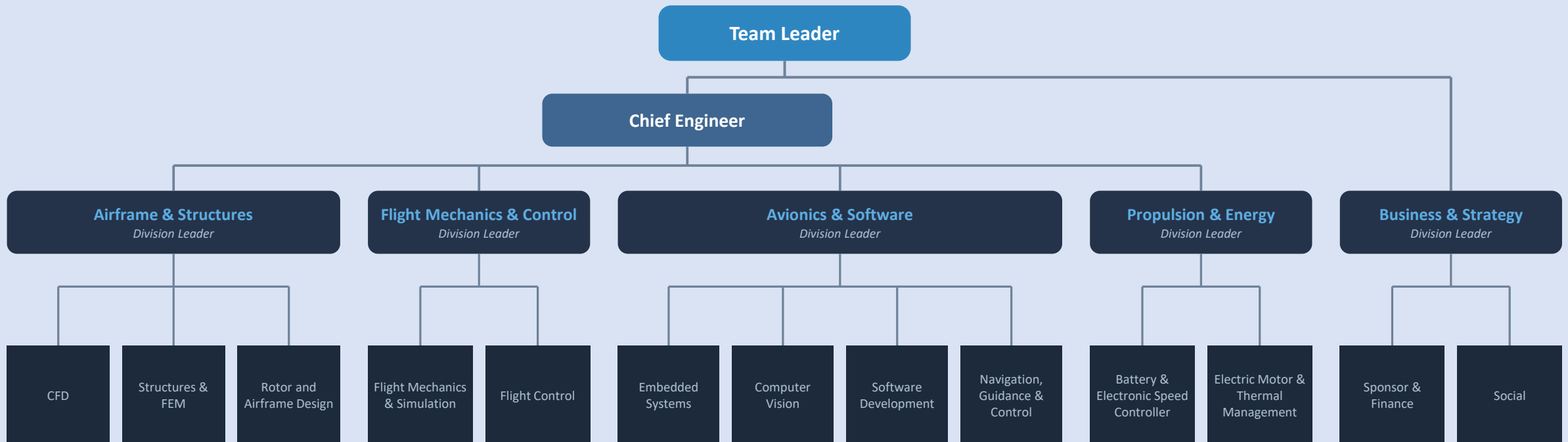


Organizational Chart

ARES operates as a lean, **company-like organization** with two clear reporting lines.

Business & Strategy, covering sponsorships, finance and outreach, reports directly to the **Team Leader**, keeping external relations close to overall decision-making.

All engineering work, instead, is coordinated by a **Chief Engineer**, who oversees the four technical divisions and ensures consistency across the aircraft's development. Each division is led by its own **Division Leader**, who manages a set of focused sub-teams responsible for specific technical areas.



International Competitions

Our competition strategy is aligned with the prototype development roadmap, allowing the team to participate in design-focused competitions from the early stages of the project, while progressively targeting competitions requiring a fully operational electric helicopter as the prototype reaches maturity.

VFS

Student Design Competition

May 31, 2027/2028
depending on project progress



VFS

Design-Build-Vertical Flight Competition

2028



2028

SUAS

Competition



Why Partner With Us?



Research & Development

Access to a student-led engineering team working across the full development cycle, from parametric design and aerodynamic and structural simulations to composite manufacturing and autonomous control systems, giving companies direct exposure to innovative engineering solutions and emerging talent.



Access to Talent

Direct access to talented Aerospace Engineering students at Politecnico di Torino. Engage with highly motivated future engineers, observe their technical and problem-solving skills in a real project environment, and identify candidates for internships, thesis projects, and employment.



Visibility & Brand

Enhanced brand visibility through presence across team materials, technical presentations, social media, and exhibition booths at international vertical-flight competitions, strengthening the company's association with innovation and the future of aerial mobility.



Let's Collaborate and become a team Partner

Let's build the future of autonomous aerial helicopters together



E-mail

ares.polito.team@gmail.com



LinkedIn

<https://www.linkedin.com/company/ares-student-team/>

Team advisors:

- **Dr. Stefano Primatesta** Fixed-term Researcher
Department of Mechanical and Aerospace Engineering (DIMEAS)
- **Prof.ssa Manuela Battipede** Confirmed Associate Professor
Department of Mechanical and Aerospace Engineering (DIMEAS)

